
SUMMARY

Business Intelligence Engineer with production SQL, ETL, and dimensional data-modeling experience turning large, multi-dimensional datasets from multiple sources into automated reporting and statistical analysis for business stakeholders. Consolidated 5M+ records across six systems into a cloud data warehouse (BigQuery, AWS S3/EMR/Athena), cut monthly close from 3-5 days to 1 day, automated 75% of recurring reporting, and built data-quality checks that catch errors before they reach a deliverable. Skilled in SQL and Python for data processing, statistical analysis (R), and visualization in Tableau. Online M.S. Computational Data Analytics, Georgia Tech, in progress.

CORE SKILLS

SQL and Data Modeling: SQL, dimensional and star-schema data modeling, data warehousing, query optimization, data quality and integrity validation

ETL and Data Engineering: ETL and pipeline design, PySpark, Docker, CI/CD

Cloud and Warehousing: AWS (S3, EMR, Athena), BigQuery, FastAPI, REST API integration (Intuit QuickBooks)

BI and Reporting: Tableau, Plotly, automated reporting, KPI definition, dashboard design

Analysis and Statistics: Statistical analysis, hypothesis testing, regression, classification, clustering, A/B testing and experimentation, causal inference (difference-in-differences, propensity score matching, IPW, AIPW), forecast backtesting

Programming: Python (pandas, scikit-learn, PyTorch, statsmodels), SQL, R

PROFESSIONAL EXPERIENCE

Starpoint Investment

Oct 2024 to Present

Data Scientist

- Architected **scalable ETL pipelines** in **BigQuery** to consolidate POS, e-commerce, inventory, and vendor invoice feeds, unifying **5M+ records** across **10+ portfolios**. Optimized transformation logic using layered CTEs and ROW_NUMBER() deduplication, and built automated anti-join integrity checks to flag orphaned records, eliminating downstream reconciliation errors to enable a **1-day financial close**.
- Engineered a **parameterized reporting layer** using advanced SQL window functions (SUM() OVER, LAG/LEAD) to automate rolling balances and period-over-period variance analysis. Replaced manual spreadsheet workflows and integrated pre-delivery data-quality validations, compressing the monthly reporting cycle from **5 days down to 1 day** while ensuring 100% deliverable accuracy.
- Automated **75%** of monthly data preparation by designing an ingestion pipeline utilizing LLMs for unstructured data extraction and validation across PDF and Excel vendor documents. Maintained strict data integrity controls, eliminating manual processing bottlenecks and saving **15+ hours per month** in manual overhead.
- Partnered with the accounting team to define reconciliation and reporting requirements, then delivered a containerized ingestion and decisioning pipeline that automates accounts-payable reconciliation across six platforms (email, DoorDash, Uber, Sysco, Shopify, Amazon), processing **~120K invoices per month** against QuickBooks via the Intuit API at an **88% straight-through** reconciliation rate. Adopted by the accounting team as their primary AP workflow.
- Built and owned the ML classifier behind the AP automation, reaching **0.925 macro-F1** for **~100 GL expense labels** on **10+ years** of client-coded history. Chose per-class thresholds with abstention over global accuracy, since misclassifying an expense as capitalizable equipment changes the client's tax treatment. A 1,000-case offline test suite gates every threshold and model change before deployment, designed to block any regression in the straight-through rate.

Merck & Co.

May 2023 to Aug 2023

Data Science Intern

- Engineered PySpark ingestion pipelines on AWS EMR landing manufacturing and instrument data into partitioned S3 tables, lowering ingestion latency **40%**, and authored reusable SQL and Python templates on Athena that cut time-to-data **60%** for four downstream analytics and engineering teams.
- Built a hybrid ML-physics forecasting model (digital twin) that uses an **RNN** to estimate physical parameters from time-series sensor data, feeding a mechanistic simulation to forecast process behavior and optimize control settings for yield and purity. Benchmarked **8** candidate models over **570K+** observations with holdout backtesting on unseen conditions plus residual and sensitivity.

Adaptive Biotechnologies

Jun 2022 to Aug 2022

Data Science Intern

- Developed an unsupervised segmentation pipeline over **100K+** single-cell samples with no labeled target: PCA at **40 components** retaining **35.5%** variance, feeding a comparison of k-means against DBSCAN to avoid assuming cluster geometry, with final structure selected on silhouette score and stability under resampling. Revealed previously unrecognized subpopulations that domain scientists validated and adopted as a new research direction, presented to department leadership.

PROJECTS

Causal Inference and Uplift Modeling, Criteo Advertising Data [GitHub](#)

- Recovered a known causal effect from deliberately broken data: **14M** ad impressions with confounding injected by a prognostic-score selection rule at **0.947** holdout AUC, pushing the naive conversion ATE to **3.4x** the RCT ground truth. Doubly-robust AIPW reduced bias from **+237% to +4.6%**, while IPW reached textbook covariate balance at all |SMD| below 0.05 yet remained **92% biased** as its effective control sample collapsed to **192K of 1.06M**. Benchmarked nine estimator configurations (PSM, IPW, AIPW, g-computation, DiD under satisfied and violated parallel trends) against placebo tests, propensity trimming, and Rosenbaum bounds.
- Ranked audiences by predicted CATE on a cluster-split holdout of **3.6M** rows with no user appearing on both sides: a transformed-outcome learner reached **Qini 0.659** and **6.6x** uplift in the top decile, while a plain response model outranked both causal learners at **Qini 0.813**, establishing that uplift modeling must be justified against that baseline. Served the selected model behind FastAPI in Docker with gated retraining on a holdout-Qini floor and PSI drift monitoring.

Constrained Budget Optimization for Marketplace Campaigns [GitHub](#)

- Formulated and evaluated a fixed-budget reallocation policy (constrained optimization): isolated low-efficiency campaign segments consuming **34.6%** of budget while producing only **22.9%** of approved conversions across **\$20.3K** spend, then modeled a reallocation policy under fixed-budget constraint, projecting **+148** incremental approved conversions and CPA improvement from **\$58.53 to \$50.98**.

EDUCATION

Georgia Institute of Technology, M.S. Computational Data Analytics

Expected Dec 2028

SUNY Buffalo, B.A. Computational and Applied Mathematics

Aug 2022 to Dec 2023